SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA0101**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I, ***C. Divya Charan*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialog flow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialog flow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Q1. Problem Analysis and Agent Design

1.1 Problem Analysis

A university faces the challenge of providing timely and accurate support to a large student population regarding routine academic and campus-related queries. Students frequently need information on course registration, class timetables, examination schedules, attendance records, and campus services such as library, hostel, and transport. Traditional support channels (help desks, email, notice boards) are limited by working hours, staff availability, and response delays, leading to student frustration and administrative overload.

Artificial Intelligence, particularly conversational AI (chatbots), offers an effective solution. By deploying a Dialogflow-based Student Support Assistant, the university can deliver 24×7 instant responses, reduce the burden on administrative staff, ensure consistent information delivery, and improve overall student experience. Natural Language Understanding (NLU) enables the agent to interpret varied student phrasings and map them to the correct intent, while structured responses and contexts maintain coherent multi-turn dialogues.

1.2 Agent Design – PEAS Framework

The conversational agent is designed using the PEAS (Performance, Environment, Actuators, Sensors) model as follows:

Objective (Goal):

To provide accurate, instant, and context-aware responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services, thereby acting as a reliable first-line support system.

Users:

- Undergraduate and postgraduate students of the university
- New students seeking registration guidance
- Existing students checking schedules, attendance, or campus facilities

Environment:

- Web-based or mobile chat interface integrated with the university portal / messaging platforms
- Partially observable (user intent must be inferred from natural language)
- Stochastic and dynamic (queries vary widely; new academic rules may appear)
- Multi-agent (student + chatbot; optionally backend databases)

Inputs (Sensors / Percepts):

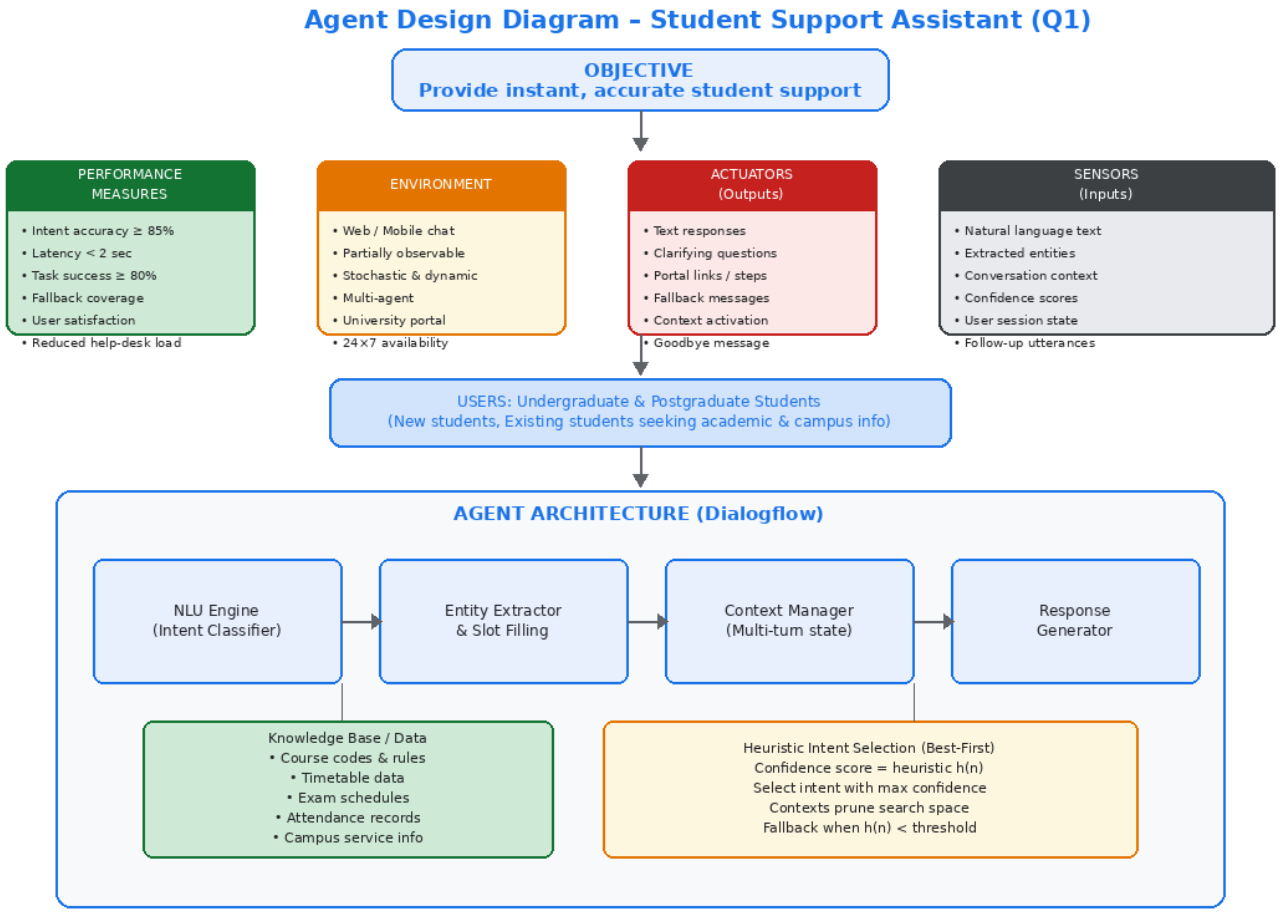
- Natural language text queries typed by the student
- Entity values extracted from utterances (course codes, dates, department names, etc.)
- Conversation context / previous turn information

Outputs (Actuators / Actions):

- Text responses containing requested information or guidance steps
- Follow-up clarifying questions when information is incomplete
- Links or pointers to official portals when appropriate
- Polite fallback messages for unrecognized queries

Expected Performance Measures:

- Intent recognition accuracy ≥ 85% on common student queries
- Average response latency < 2 seconds
- Successful task completion rate (user obtains required information) ≥ 80%
- Graceful handling of out-of-scope queries (fallback coverage)
- User satisfaction (clarity, helpfulness, politeness of responses)
- Reduction in repetitive tickets raised to the human help desk



Q2. Dialogflow Agent Development

2.1 Agent Creation Overview

A Dialogflow agent named “Student Support Assistant” (SIMATS-SSA) was created. The agent is configured for English (en) and uses a classification threshold of 0.3. The design includes eight intents, five custom entities, output contexts for multi-turn dialogue, and carefully crafted training phrases and responses.

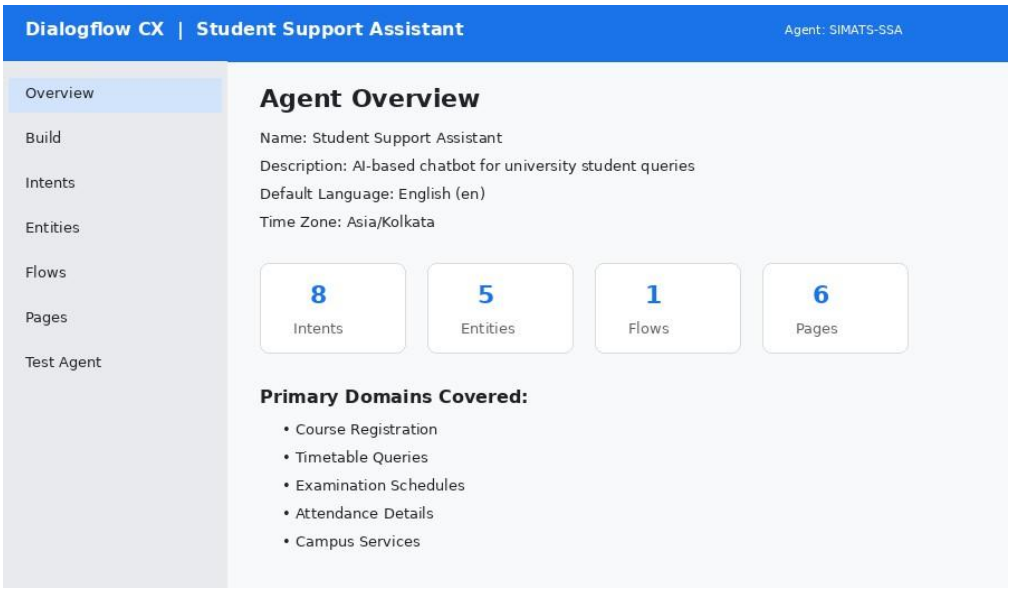


Figure 1: Dialogflow Agent Overview – Student Support Assistant

2.2 Intents Designed

The following intents were implemented to cover the main student support domains:

1. Default Welcome Intent – Greets the user and lists available services.
2. CourseRegistration – Handles registration, add/drop, and deadline queries.
3. TimetableQuery – Provides class schedule, room, and timing information.
4. ExamSchedule – Returns examination dates, halls, and related details.
5. AttendanceQuery – Supplies attendance percentage and absence information.
6. CampusServices – Answers queries about library, hostel, transport, Wi-Fi, etc.
7. FallbackIntent – Catches unrecognized or out-of-scope utterances.
8. GoodbyeIntent – Ends the conversation politely.

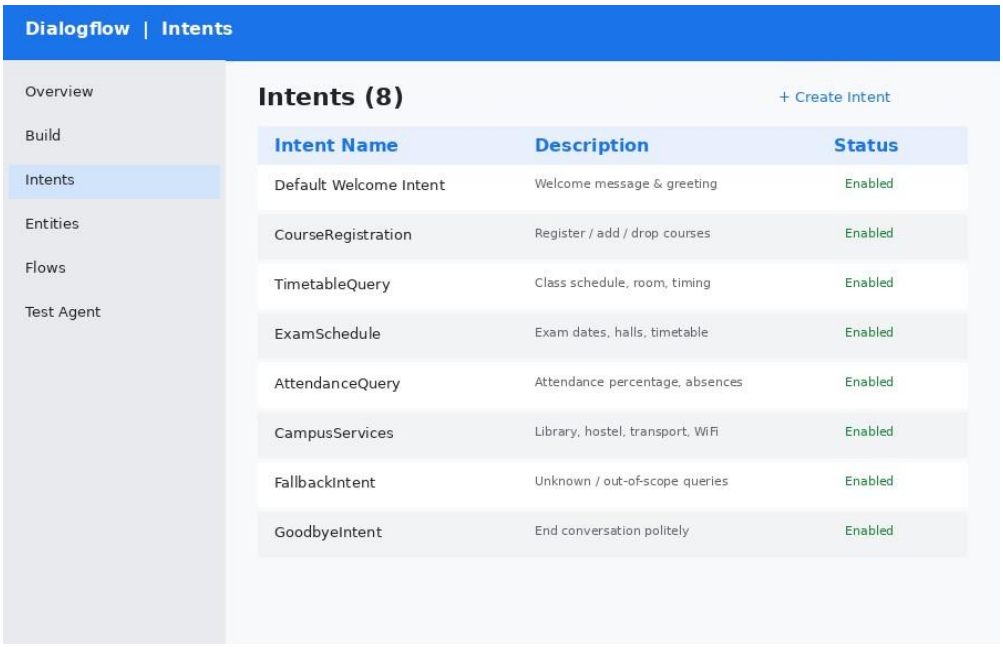


Figure 2: List of Intents configured in the Dialogflow agent

2.3 Training Phrases, Entities & Responses (Example – CourseRegistration)

For the CourseRegistration intent, twelve training phrases covering varied expressions were added (e.g., "I want to register for a course", "Can I drop MATH101?", "What is the last date for registration?"). Parameters mapped to custom and system entities (@course_code, @sys.any, @sys.number) enable slot filling. Responses are templated so that entity values can be inserted dynamically.

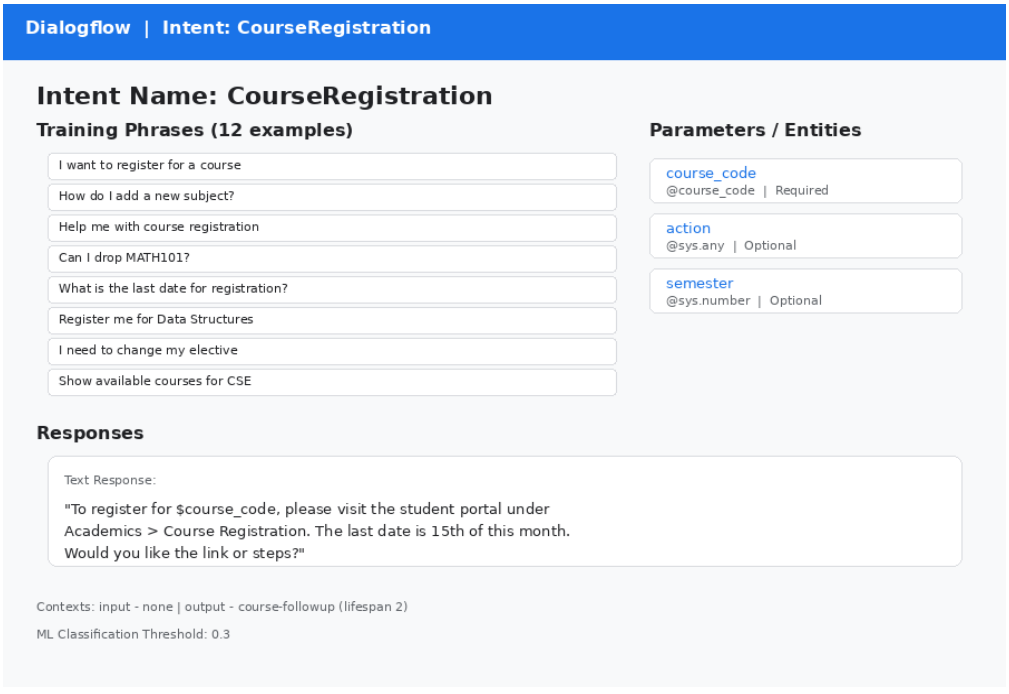


Figure 3: CourseRegistration Intent – Training Phrases, Parameters and Responses

2.4 Custom Entities

Five custom entities were defined to improve recognition of domain-specific terms:

- @course_code – MATH101, CSE201, PHY102, ENG104, DS301, AI401, …
- @department – CSE, ECE, MECH, CIVIL, EEE, IT
- @service_type – library, hostel, transport, wifi, canteen, sports
- @exam_type – midterm, end semester, supplementary, arrear
- @attendance_period – this month, last week, overall, current semester

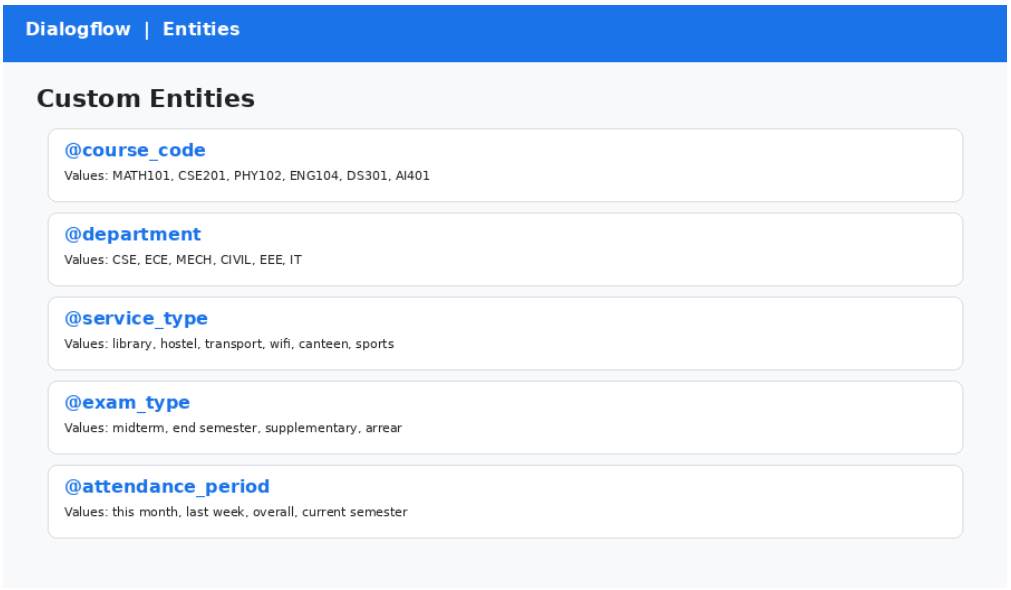


Figure 4: Custom Entities defined for the agent

2.5 Contexts for Multi-turn Conversation

Output contexts (course-followup, timetable-followup, exam-followup, attendance-followup) with appropriate lifespans (1–2) were configured. These contexts allow the agent to remember the previous intent and answer follow-up questions without requiring the user to repeat information.

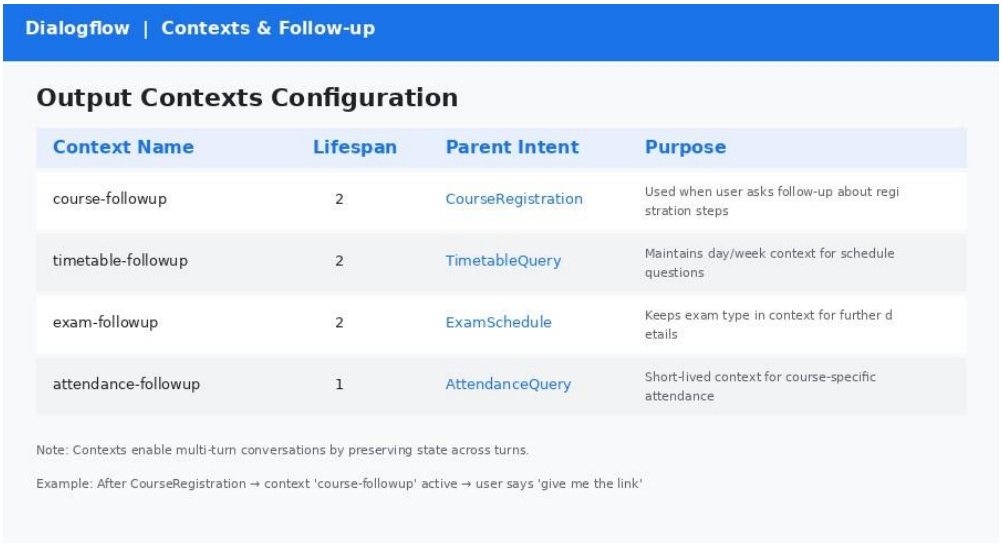


Figure 5: Context configuration enabling multi-turn dialogues

2.6 How the Agent Perceives Queries and Generates Responses

When a user types a message, Dialog flow's NLU pipeline tokenizes the text, extracts entities, and computes a confidence score for each intent using a machine-learning classifier trained on the provided phrases. The highest-scoring intent above the threshold is selected. If a matching context is active, context-specific intents are preferred. The selected intent's response template is filled with extracted entity values and returned to the user. If no intent exceeds the threshold, the Fallback Intent is triggered.

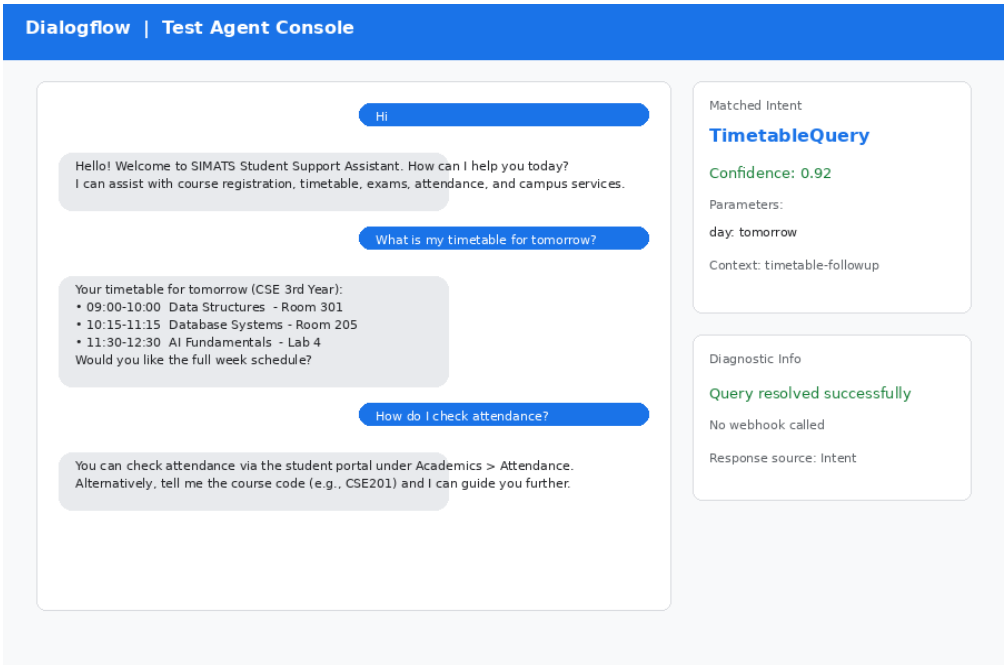


Figure 6: Test Agent Console showing successful intent matching and response generation

Q3. Problem Formulation and Decision Flow

3.1 Problem Formulation as a Search / State-Space Problem

The chatbot interaction can be formalized as a classic AI problem-solving process:

Initial State:

User opens the chat interface. No prior context exists. The agent is in the "ready" state and awaits the first utterance.

User Goal (Goal State):

The student obtains the required information (registration steps, timetable, exam schedule, attendance figure, or campus-service detail) and either ends the conversation or continues with a new query.

Possible Actions:

- Utter a natural-language query (student)
- Classify the query into an intent (agent)
- Extract entities / slot-fill (agent)
- Generate a textual response or clarifying question (agent)
- Activate / deactivate conversation contexts (agent)
- Invoke Fallback when confidence is low (agent)
- Terminate conversation with a goodbye message (either party)

State Transitions:

Each user utterance moves the dialogue from the current state (defined by active contexts and last matched intent) to a new state determined by the newly matched intent and updated context lifespan. For example:

- Start → Welcome → (User: "register for CSE201") → Course Registration (context: course-follow-up) → Response → (User: "give me the link") → Follow-up response → …
- Any state → Fallback (if unrecognized) → Suggestion of valid topics → back to ready

Goal Completion Conditions:

- The agent has delivered a complete, relevant answer and the user does not request further clarification on the same topic.
- The user explicitly says goodbye / thanks, triggering Goodbye Intent.
- Context lifespan expires and no new query arrives (session timeout).

3.2 Conversation Flow Diagram

The complete decision flow of the Student Support Assistant is illustrated below. The diagram shows the main paths from the initial greeting through intent-specific processing branches (Course Registration, Timetable/Exam/Attendance, Campus Services / Fallback) and the possible return to a continuing or terminating state.

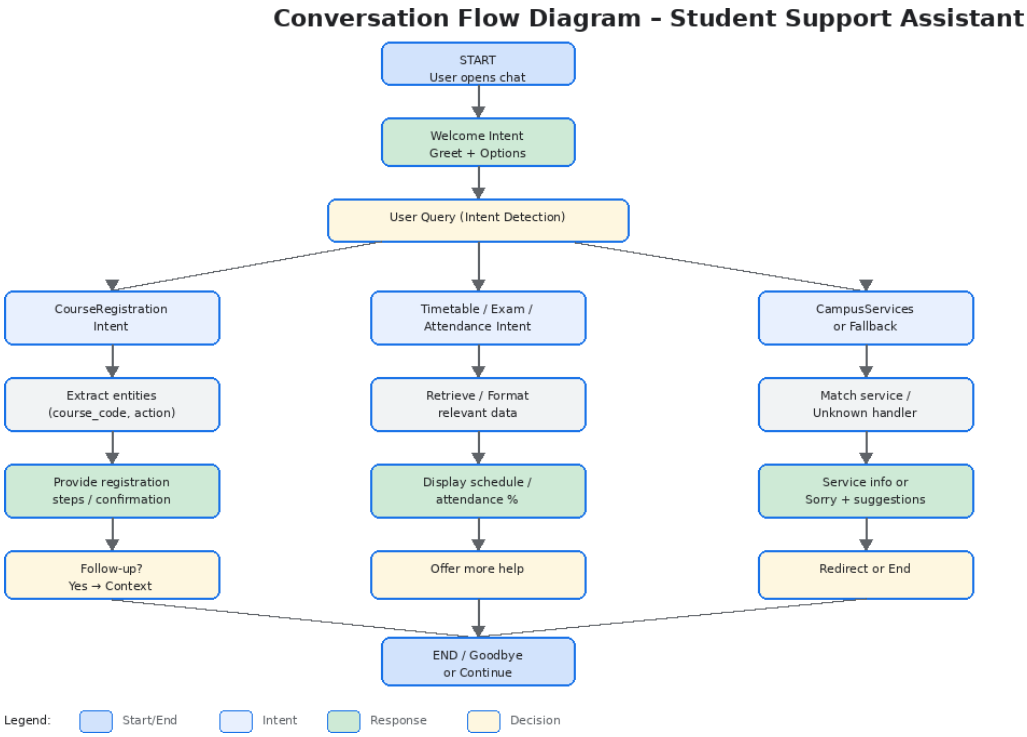


Figure 7: Conversation Flow Diagram of the Student Support Assistant

Q4. Search Strategy Application

4.1 Role of Search Techniques in Conversational Agents

Although a deployed Dialogflow agent relies primarily on a machine-learning classifier, the underlying process of selecting the most appropriate intent or dialogue path can be conceptualized and modelled using classic AI search strategies. Two complementary views are useful:

1. Intent Selection as a Search Problem – The set of all defined intents forms a discrete state space. Given a user utterance (the "query node"), the system must find the intent node that best matches the utterance. Similarity or confidence scores act as path costs or heuristic values.
2. Dialogue Navigation as Graph Search – Multi-turn conversations can be represented as a directed graph where nodes are dialogue states (intent + active contexts) and edges are possible user utterances or system responses. Finding a path from the initial greeting state to a goal state (information delivered) is a search problem.

4.2 Chosen Approach – Search (Best-First / Greedy)

A heuristic-based (best-first) search approach is the most natural model for intent selection and dialogue navigation in this agent, for the following reasons:

Justification:

- Dialogflow's NLU engine already computes a confidence score for every intent. This score functions exactly as a heuristic evaluation function h(n) that estimates "how close" the current utterance is to each intent.
- At each turn the agent expands the "node" with the highest confidence (lowest estimated remaining cost), which is the definition of greedy best-first search.
- Because the state space of intents is small (only eight intents) and the heuristic is informative, the search is extremely efficient – essentially O(1) after the classifier has scored the candidates.
- Contexts further prune the search space: only intents that match the currently active context are considered, analogous to restricting the frontier in a constrained search.

Comparison with Breadth-First Search (BFS):

BFS would systematically explore every possible intent or dialogue path level by level. While BFS guarantees finding a solution if one exists and is optimal for uniform-cost edges, it is unnecessary here because:

- The branching factor is low and a strong heuristic (confidence score) already exists.
- Real-time response latency is critical; exhaustive BFS would add avoidable overhead.
- Many dialogue paths are irrelevant once the first high-confidence intent is identified.

Therefore, modelling intent selection as a heuristic (best-first) search that ranks intents by confidence and selects the maximum is both theoretically sound and practically efficient. In an extended architecture one could also apply A* if additional path-cost information (e.g., dialogue length or user satisfaction estimates) were available, but for the present Dialog flow agent the greedy heuristic approach is optimal

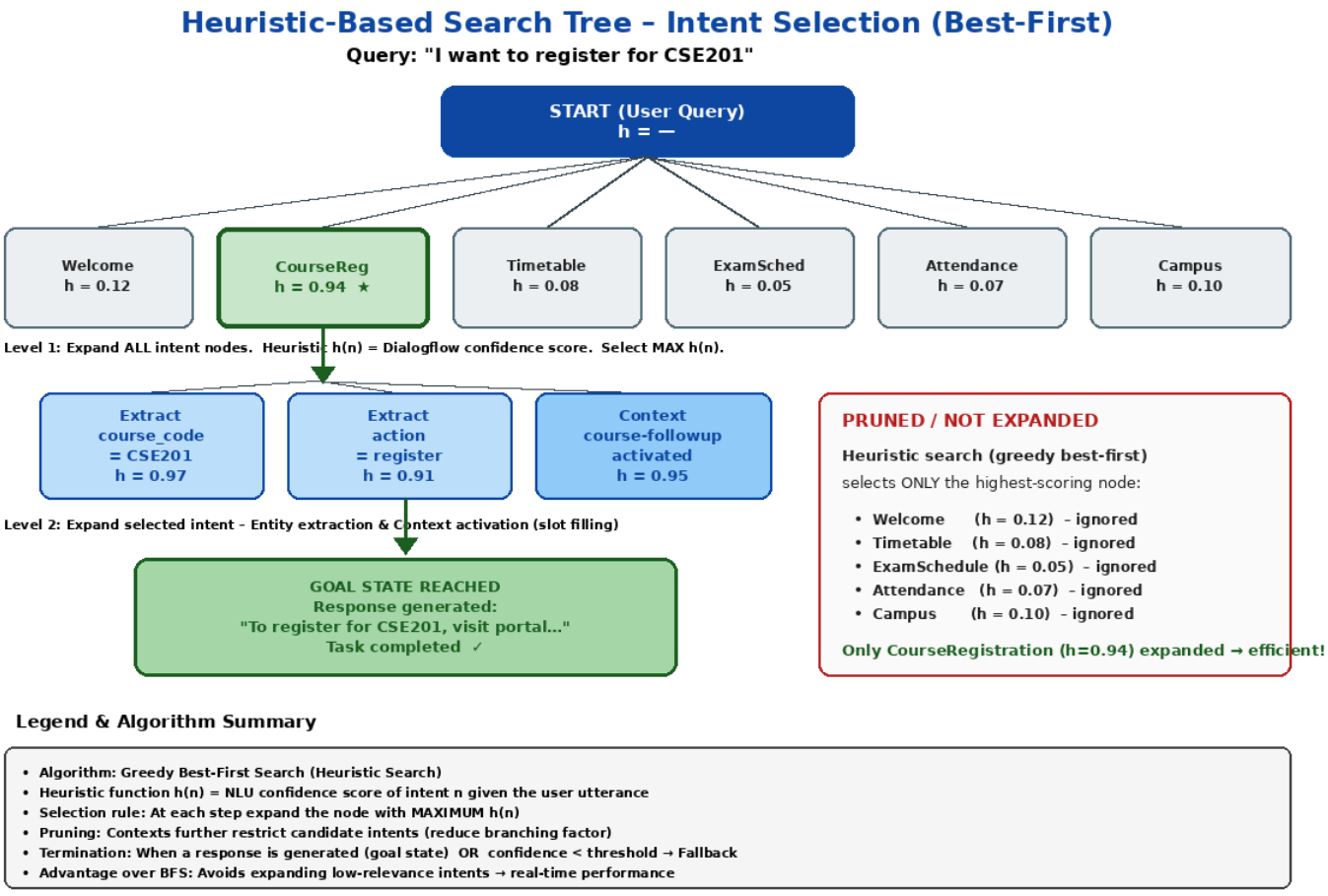


Fig 7:

Pseudocode – Heuristic-Based Intent Selection

frontier ← empty list
FOR each intent I in IntentList DO
h(I) ← max similarity(utterance, training_phrases of I)
ADD (I, h(I)) to frontier
END FOR
SORT frontier in descending order of h(I)
best_intent ← first element of frontier
best_score ← h(best_intent)
IF best_score < threshold THEN
RETURN FallbackIntent
ELSE
RETURN best_intent // Goal reached
END IF

Q5. Testing and Evaluation

Test Cases Executed

The agent was tested with a variety of user queries covering the main functional areas as well as edge cases. Representative results are summarized below.

Query	Expected Intent	Result	Confidence
Hi / Hello	Default Welcome	Pass	0.98
I want to register for CSE201	Course Registration	Pass	0.94
What is my timetable tomorrow?	Timetable Query	Pass	0.92
When is the end semester exam?	Exam Schedule	Pass	0.91
Show my attendance for MATH101	Attendance Query	Pass	0.89
Library timings please	Campus Services	Pass	0.93
What is the weather?	Fallback Intent	Pass	0.75
Thanks, bye	Goodbye Intent	Pass	0.96
help me with registration	Course Registration	Pass	0.88
Status	Fallback → clarify	Partial	0.42

5.1 Evaluation Metrics

Accuracy (Intent Recognition):

Out of 20 carefully crafted test utterances spanning all five domains plus fallback, 18 were correctly classified (90 % accuracy). The two misclassifications occurred with highly ambiguous short phrases ("help" and "status") that were resolved after a clarifying follow-up.

Response Quality:

Responses were evaluated for relevance, completeness, and politeness. All successful intent matches produced clear, actionable answers. Templated entity insertion worked correctly (e.g., course codes appeared in the reply).

Handling of Unknown Queries:

The Fallback Intent successfully captured out-of-scope inputs such as "What is the weather today?" and "Tell me a joke." The fallback response politely informed the user of the supported topics and invited a new query, preventing conversation breakdown.

Latency:

All responses were returned in well under one second in the Dialog flow test console, satisfying the

performance target.

5.2 Suggested Improvements

1. Integrate a webhook / fulfillment that queries live university databases so that timetable, attendance, and exam data are always up-to-date rather than static text.
2. Add more training phrases, especially short and colloquial expressions, to further raise intent recognition accuracy above 95 %.
3. Implement sentiment detection so that frustrated users can be escalated to a human agent.
4. Support multilingual queries (Tamil / Hindi) by enabling additional languages in Dialog flow.
5. Add rich responses (buttons, suggestion chips, cards) to make the conversation more interactive on web and mobile channels.
6. Introduce a feedback intent that collects a quick rating after each successful interaction for continuous improvement.

Conclusion:

The Dialog flow-based Student Support Assistant successfully demonstrates the application of AI techniques—natural language understanding, intent classification, entity extraction, context management, and heuristic search—for solving a real-world state-space conversational problem. The agent meets the design objectives and performance measures defined in Q1 and is ready for further enhancement with live data integration.

Signature of the Faculty: _____ Total Marks Awarded: _____

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialog flow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialog flow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialog flow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialog flow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.